

Study of stable isotopes and their applications in the marine environment

OCEA0027-A-a



Photo: Mare aux Chevreuils du Sart-Tilman

**Study of the trophic
position of gastropods
from ponds 1 and 2**

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Background

As part of the practical session:

- Sampling in two ponds
 - Distinct environments
 - Distinct ecosystems

Collected samples :

- Filters
- Primary producers
- Consumers
 - Bivalves, gastropods, isopods, amphipods, ...

Isotopic analysis :

- $\delta^{15}\text{N}$
- $\delta^{13}\text{C}$

Do gastropods from ponds 1 and 2 occupy the same trophic position?

Gastropods :

- Varied diet :
 - Herbivores / detritivores
 - Algae, plants, biofilms
 - Carnivores
 - Aquatic organisms or gastropods
- Crucial role in the ecosystem :
 - Recycling of organic matter
 - Control of algal populations
 - Food source for other animals:
 - amphibians, birds, aquatic insects...



Common ramshorn snail
(*Planorbis planorbis*)



Materials and Methods :

METHODOLOGY :

- Sampling of organisms

Pond 1 :

- Forest
- Stream
- Sulfur odor

Pond 2 :

- Plain
- Road
- Fields

- Creation of an Excel file
- Preparation of organisms
- Isotopic analyses
- Study of the results

DATA ANALYSIS :

- Data observation
- ANOVA :
15
○ δ N ~ ponds -
 - Primary producers
 - Consumers
 - Gastropods
- Trophic position test using a model
 - Gastropods
 - Comparison with other species

Results and discussion :

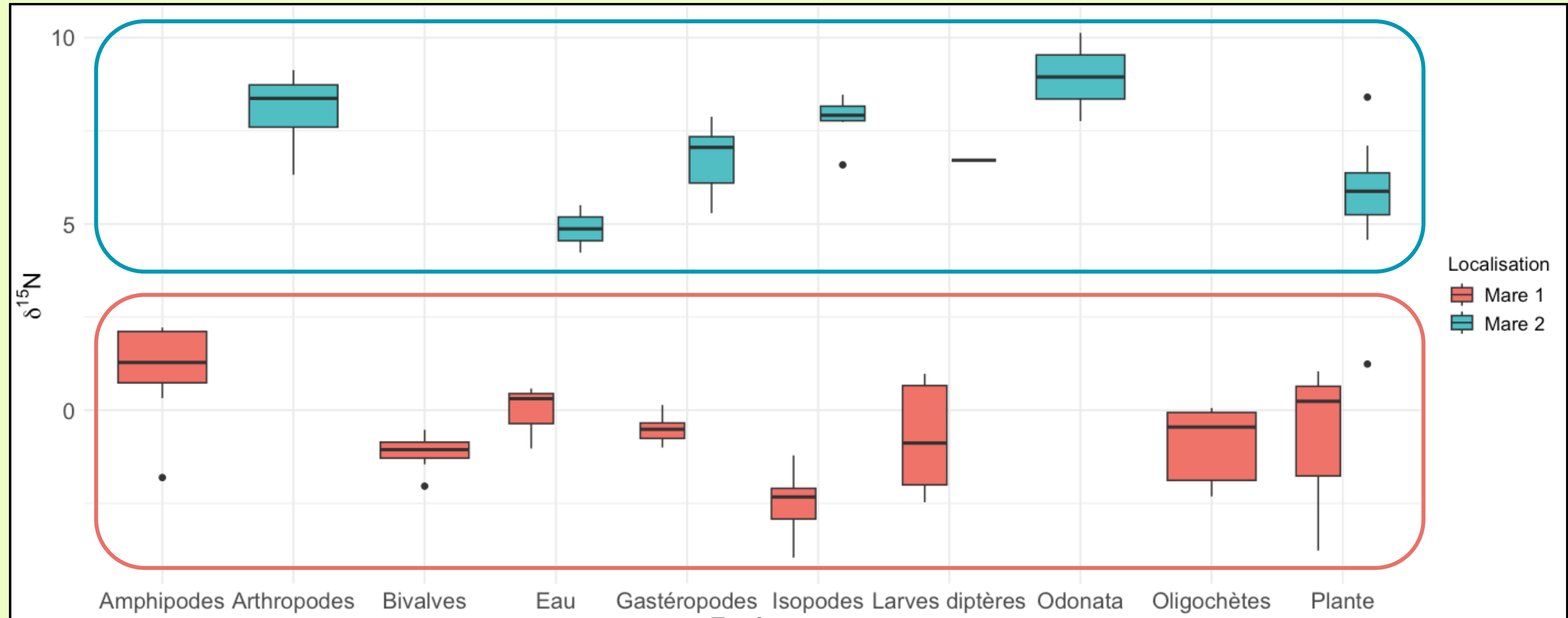


Figure 1: Variation of $\delta^{15}\text{N}$ by species and by site

POND 1 :

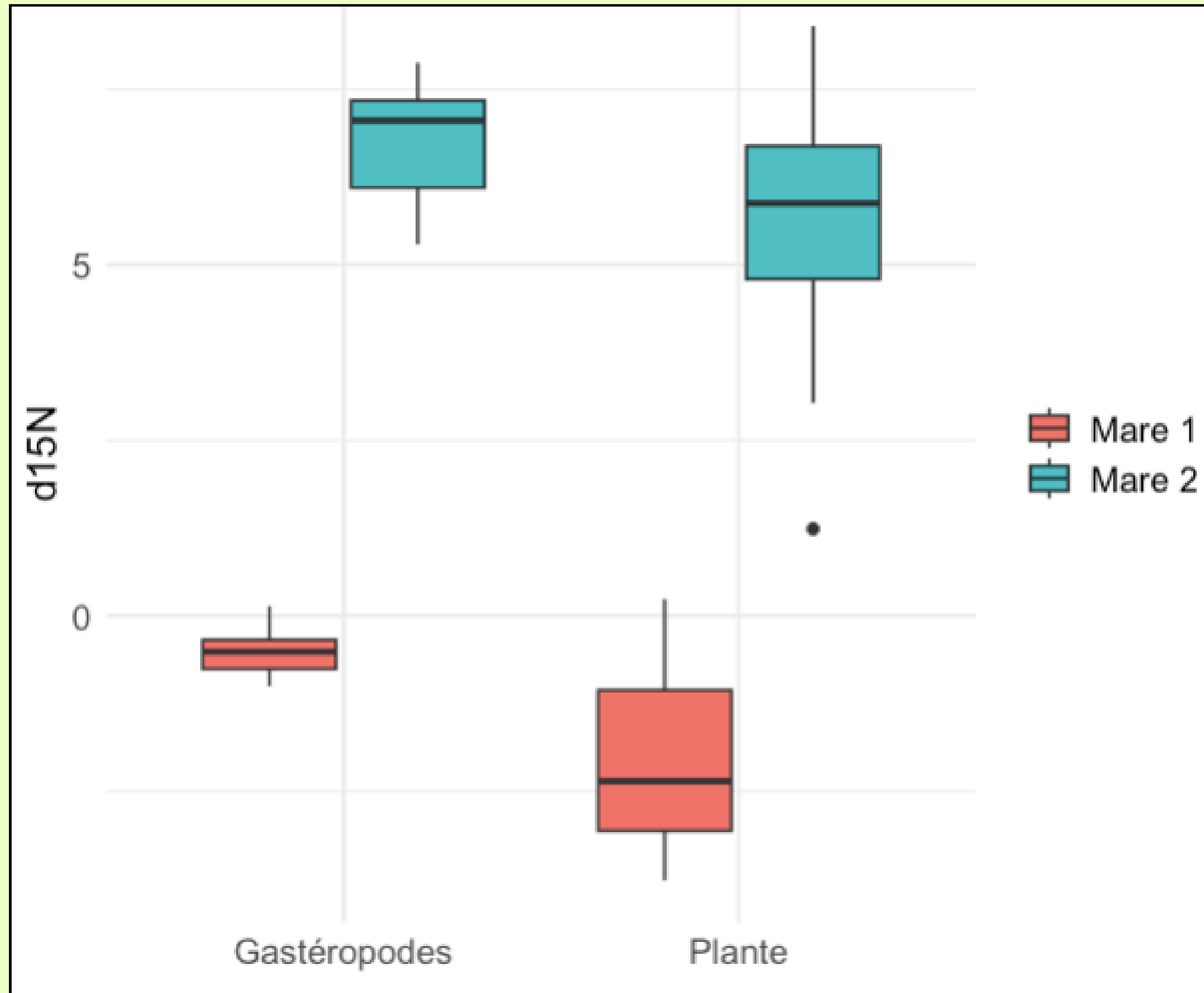
- Negative values
 - Anoxic environment
 - Specific isotopic fractionation

POND 2 :

- Higher values
 - Anthropogenic inputs
 - Frequently ^{15}N -enriched



Results and discussion :

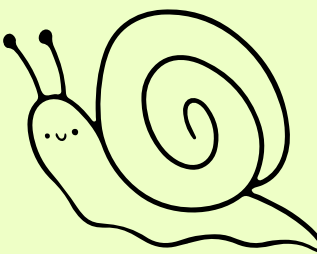


ANOVA :

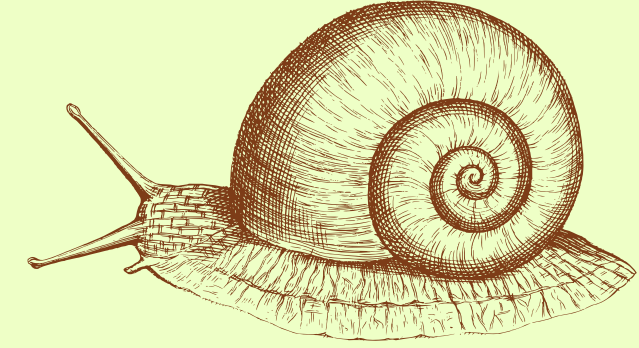
Significant differences in $\delta^{15}\text{N}$ values between ponds:

- Gastropods
- Producers
- Consumers

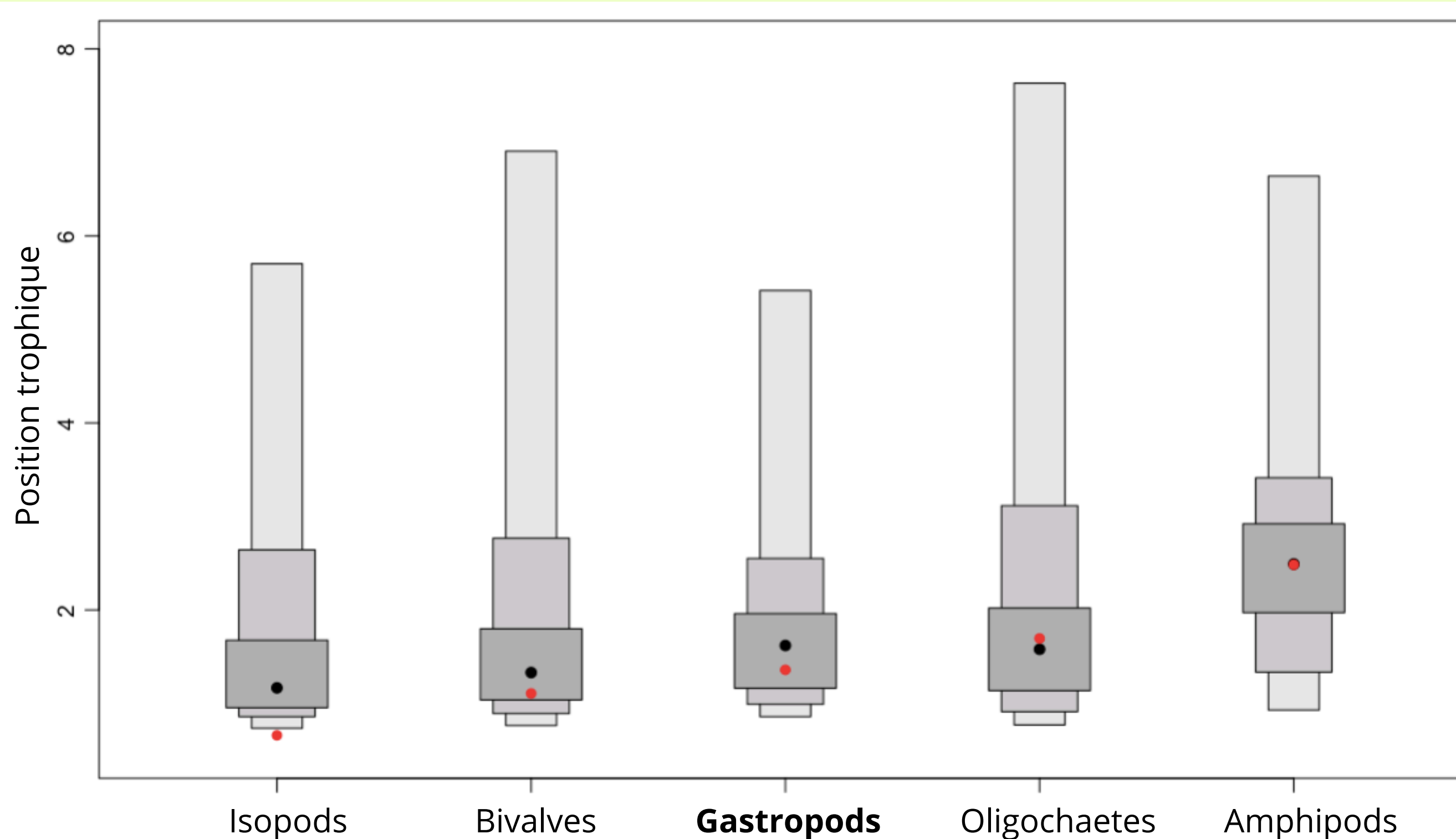
Figure 2: Variation of $\delta^{15}\text{N}$ for gastropods and plants across sites



Results and discussion :



Trophic positions in Pond 1:



- Isopods TP = 0.66
 - Detritivores
- Bivalves TP = 1.11
 - Filter-feeding and herbivores
- **Gastropods TP = 1.46**
 - **Herbivorous**
 - **or detritivores**
- Oligochaetes TP = 1.7
 - Detritivores
- Amphipods TP = 2.48
 - Omnivores

Figure 3: Trophic position of different species in Pond 1

Results and discussion :

Trophic positions in Pond 2:

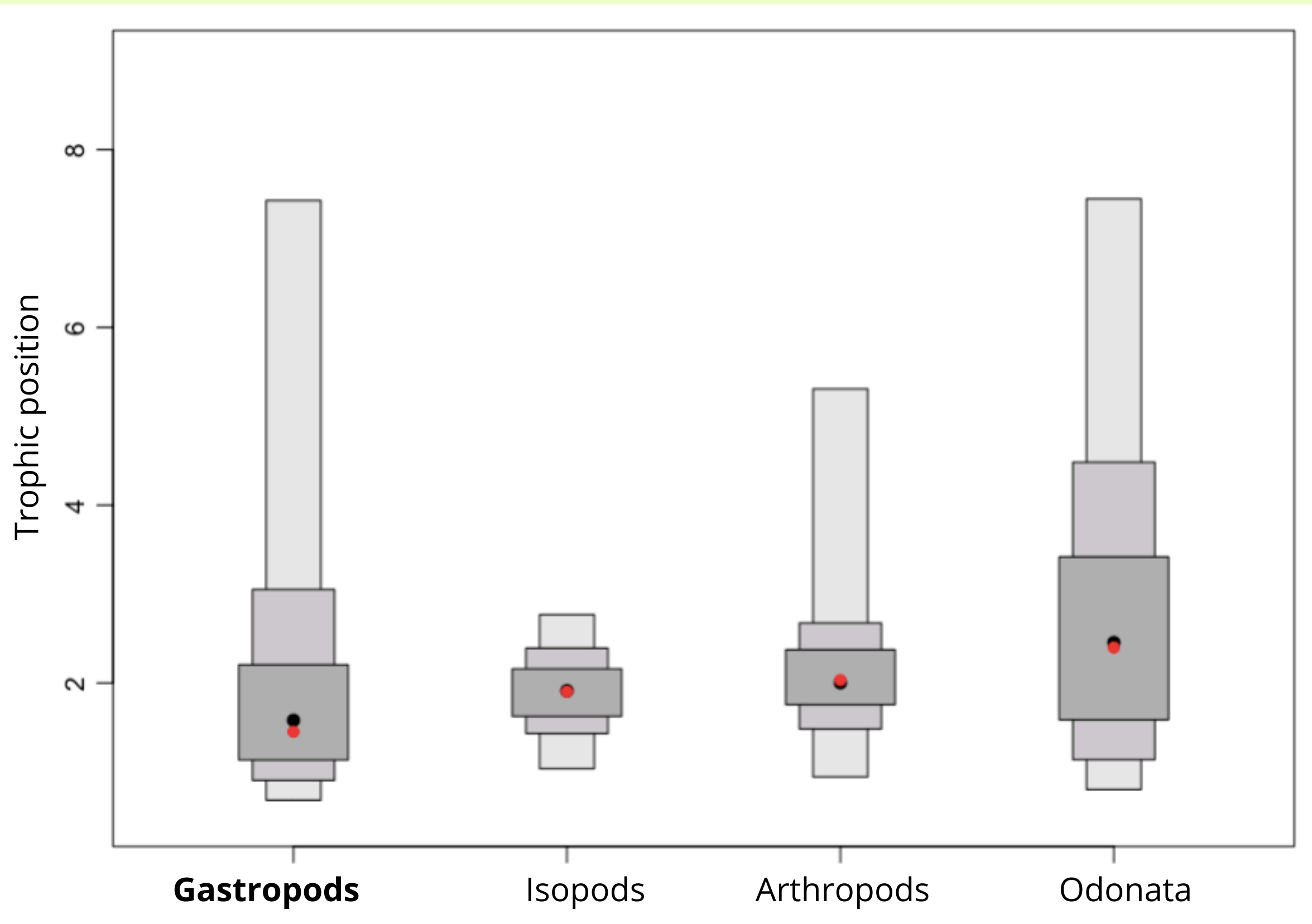
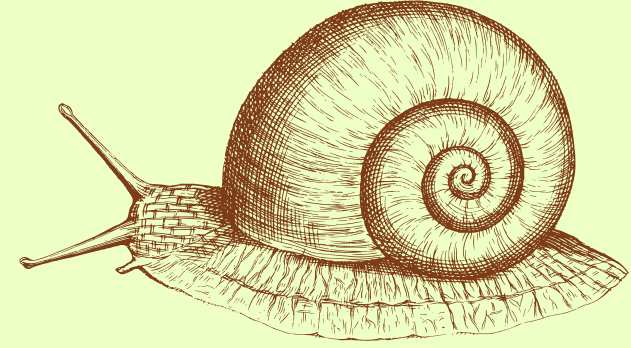


Figure 4: Trophic position of different species in Pond 2

- **Gastropods TP = 1.57**
 - **Herbivores ou détritivore**
- Isopodes TP = 1.9
 - Omnivores ou detritivores
- Arthropods TP = 2.03
 - Omnivores
- Odonata TP = 2.4
 - Carnivores

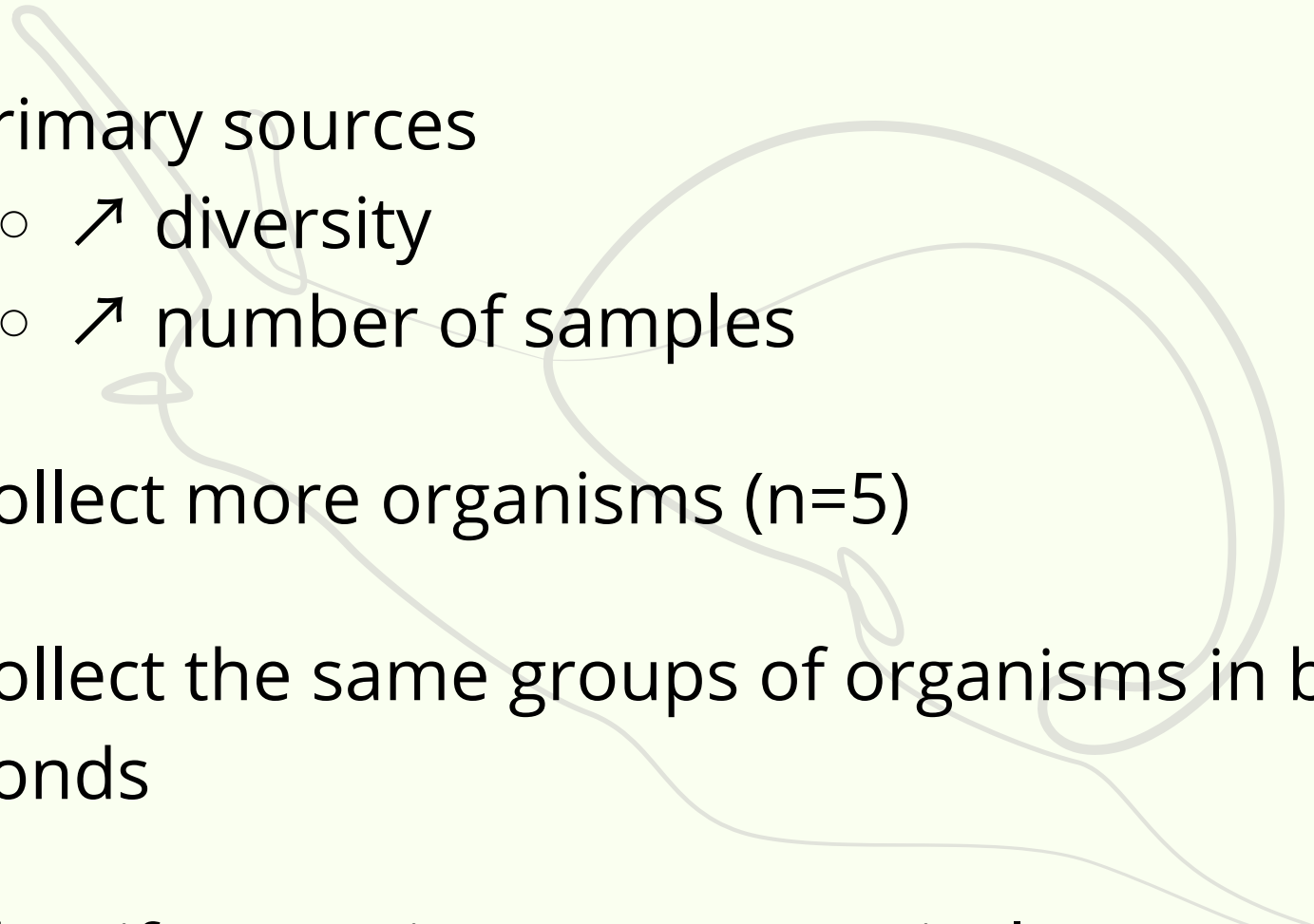
Gastropods from Ponds 1 and 2 appear to occupy similar trophic positions; they are herbivorous and/or detritivorous.

Conclusion and perspectives :

- Lower $\delta^{15}\text{N}$ in Pond 1
 - Anoxic environment
- Higher $\delta^{15}\text{N}$ in Pond 2''
 - Anthropogenic pollution
 - Road runoff + fertilizers

- Trophic positions
 - Similar between Ponds 1 and 2
 - Herbivorous / detritivorous
 - Consistent with other organisms

Further research :

- Primary sources
 - ↗ diversity
 - ↗ number of samples
 - Collect more organisms (n=5)
 - Collect the same groups of organisms in both ponds
 - Identify organisms more precisely
- 

A close-up photograph of a snail with a brown, ribbed shell and a white, translucent body. The snail is positioned on a highly reflective surface, likely water, which creates a clear reflection of the snail and its shell. The background is a blurred, greenish-brown, suggesting an outdoor environment. A semi-transparent white rectangular box with rounded corners is overlaid on the image, containing the text.

Thank you for your attention !

Do you have any questions ?